Dr. Abderraouf Maoudj

Robot learning · Multi-robot coordination · Resilient robotic systems

Postdoctoral Research Fellow & Coordinator of the Robotics Laboratory

Interdisciplinary Research Center for Intelligent Manufacturing and Robotics
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1,837 citations · h-index 14 · i10-index 16 (Google Scholar, November 2026)

Research Profile

Robotics researcher with twelve years of academic and industrial R&D experience in Algeria, Denmark and Saudi Arabia. Develops learning and control methods for manipulation, decentralized robot teams and fault-tolerant locomotion, supported by physical prototypes, industrial collaborations and open research infrastructure.

- **Learning for manipulation:** multi-arm reinforcement learning, imitation learning and world models; established a shared dual-arm Meca500 testbed as Principal Investigator of KFUPM Grant INMR2403.
- **Scalable robot teams:** decentralized planning, scheduling and coordination for warehouse fleets, including planners evaluated with at least 1,000 simulated robots and lifelong task streams.
- **Resilient embodied intelligence:** open-source legged platforms, learning-augmented CPG control and sim-to-real locomotion under progressive hardware failures; ongoing cross-embodiment learning through NeuroAct.

Research Appointments

2023–Present **Postdoctoral Research Fellow & Robotics Laboratory Coordinator**
IRC-IMR, King Fahd University of Petroleum & Minerals, Saudi Arabia
Learning-based manipulation and multi-arm planning; decentralized multi-agent learning; resilient legged control and cross-embodiment policies. PI, INMR2403; Co-PI, INMR2632. Graduate teaching and student supervision.

2021–2023 **Postdoctoral Researcher**
SDU Biorobotics, University of Southern Denmark
Decentralized multi-robot coordination, communication-constrained path finding and capacitated fleet scheduling for industrial logistics. Worked with Enabled Robotics on MiR–UR5 mobile manipulators; guest lecturing and Master's supervision.

2018–2020 **Senior Researcher, Robotics Division**
Centre de Développement des Technologies Avancées (CDTA), Algeria
AGV fleet control, intelligent scheduling, collaborative robot integration and Siemens PLC automation for Industry 4.0; engineering teaching and project supervision.

2014–2018 **Robotics Engineer, Robotics Division**
CDTA, Algeria
Robot development, including an industrial AGV and a remotely operated bathymetric survey boat; doctoral research and laboratory teaching at USTHB.

Education

2013–2018 **Ph.D., Process Control and Interactive Rendering**
Université des Sciences et de la Technologie Houari Boumediene (USTHB), Algeria
Distinction: *Très Honorable*. Dissertation: *Teleoperation of Heterogeneous Robots: Application to RobuTER/ULM Robots and AGVM6 Robots.* Advisor: Prof. Brahim Bouzouia.

2010–2012 **M.Sc., Electronics — System Analysis**
Mohammed-Seddik Benyahia University of Jijel, Algeria
Thesis: *Design and Control of a Microcontroller-Based Mobile Robot Prototype.*

2007–2010 **B.Sc., Electronics**
Mohammed-Seddik Benyahia University of Jijel, Algeria

Teaching, Supervision and Mentorship

Current graduate teaching

ICS 520 — Artificial Intelligence and Machine Learning for Robotics, KFUPM, Fall 2026. Autonomous drone applications spanning perception, planning, decision-making and learning.

Courses taught

- **Automation Laboratory (AUTO503)**, undergraduate, USTHB, 2014/2015 and 2015/2016; **Control Laboratory**, Master's in Electronics of Embedded Systems, 2017/2018; **Industrial Programmable Logic Controllers (API)**, Master's in Instrumentation, 2018/2019.
- **Further teaching:** Linear Control Systems; Introduction to Robotic Systems and ROS; Applied Reinforcement Learning for Robotic Manipulation; Embedded Systems for Automotive Engineering; Algorithms and C Programming; guest lectures at the University of Southern Denmark.
- **Curriculum development:** graduate courses in Reinforcement Learning for Robotics and Sim-to-Real Embodied Robotics, integrating classical control, robot learning and projects on open robotic platforms.

Graduate and project supervision

Co-supervised doctoral students at KFUPM and supervised or co-supervised Master's theses and final-year engineering projects at KFUPM, SDU and CDTA/USTHB. Topics include manipulation, industrial automation, navigation, multi-robot coordination and legged locomotion, with outcomes in peer-reviewed publications, industrial demonstrations and working prototypes.

Research Funding and Leadership

- **Principal Investigator — KFUPM Grant INMR2403 (2023–2025).** *Advancing Robotics for Industry 4.0: Integrating Collaborative Robot Systems to Enhance Industrial Tasks.* Led the project and established the shared dual-arm Meca500 skill-learning testbed. Project record.
- **Co-Principal Investigator — KFUPM Grant INMR2632 (ongoing).** *Intelligent Bipedal Robotic Platform.* Open, modular hardware and control infrastructure for locomotion research.
- **Postdoctoral Researcher — Swarm Robotics for Industry 4.0 (2021–2023).** Independent Research Fund Denmark project at SDU, with Enabled Robotics and IRIDIA, Université libre de Bruxelles. Project record.

Research and Development Projects

Next-Generation Robotic Manipulation and Learning Systems

KFUPM · Ongoing · Postdoctoral Research Fellow

Learning-based manipulation for flexible industrial automation, combining reinforcement learning, imitation learning and inter-arm attention. A graph-attention SAC policy trained with up to four arms coordinates up to ten in simulation and runs on physical Meca500 arms. Current work explores multi-agent world models for sparse-reward coordination and vision–language–action policies for manipulation.

Laboratory demonstrations · Multi-arm planning video

NeuroAct: Transferable Robot Intelligence Across Bodies and Tasks

KFUPM · Ongoing · Postdoctoral Research Fellow

A modular control stack with an embodiment representation, hierarchical policies and persistent memory, designed to reduce retraining for new robots and tasks. A shared model has generalized to unseen synthetic robot morphologies in simulation; physical deployment is the next stage.

Resilient Legged Locomotion Under Progressive Hardware Failures

KFUPM · Ongoing · Postdoctoral Research Fellow

CPG dynamics modulated by reinforcement learning sustain locomotion as multiple joints and legs fail. A single policy trained with randomized fault injection was deployed zero-shot on a physical 18-DoF hexapod, supporting research toward dependable inspection robots. Companion work couples terrain perception and gait adaptation in quadruped simulation.

SPINDER demonstration · Open platform and simulator

Intelligent Bipedal Robotic Platform

KFUPM · Grant INMR2632 · Ongoing · Co-Principal Investigator

Open, modular hardware for reproducible locomotion, balance recovery, jumping and obstacle-negotiation experiments. Replaceable mechanical, sensing and control modules, standardized interfaces and synchronized logging support quantitative comparison of classical and learning-based controllers.

Collaborative Robotics for Industry 4.0

KFUPM · Grant INMR2403 · 2023–2025 · Principal Investigator

Developed multi-cobot skill learning, coordination and motion planning for flexible production, alongside fleet coordination for storage, transport and shipment preparation. Delivered a dual-arm Meca500 testbed with an integrated ROS stack, now used as shared research and teaching infrastructure.

Project record

Swarm Robotics for Industry 4.0

SDU Biorobotics · 2021–2023 · Postdoctoral Researcher

Decentralized planning and scheduling for warehouse fleets operating with limited communication. Delivered capacitated multi-AGV scheduling methods, planners for at least 1,000 simulated robots, and lifelong planning for continuous task arrivals. Co-authored heatmap-guided learning was evaluated with up to 3,500 simulated agents; open planners and datasets support reproducible industrial-logistics research.

Project · ANTS planner · Benchmark dataset · Video

Industry 4.0 Platform Development

CDTA, with Siemens Algeria · 2016–2020 · Robotics Engineer / Senior Researcher

Integrated KUKA LBR iiwa collaborative assembly, RFID part tracking, Siemens PLCs and AGV navigation into a cyber-physical manufacturing platform, connecting robot autonomy with industrial production control.

System demonstration

Aquatic Robot for Bathymetry

CDTA · 2016–2018 · Robotics Engineer

Designed the electronics and control of a remotely operated survey boat delivered to the Algerian National Hydrological Agency, supporting bathymetric measurement and environmental monitoring.

AquaRob publication, IEEE EDiS 2024

Remote Control of Heterogeneous Robots

CDTA · 2014–2016 · Robotics Engineer / Doctoral Researcher

Distributed scheduling and control for heterogeneous industrial robots, reinforcement-learning path optimization and fault-tolerant formation control, supporting coordinated operation of mobile and manipulator teams.

Robots and Systems Designed and Built

- **SPINDER, open-source 18-DoF hexapod:** mechanical design, electronics, firmware, analytic inverse kinematics and hierarchical CPG control. Designs, models and code · Demonstration.
- **Dual-arm Meca500 testbed:** hardware integration and ROS stack for manipulation research and teaching.
- **Industrial AGV:** microcontroller-based motion control and lidar integration.
- **Industry 4.0 assembly cell:** KUKA LBR iiwa, Siemens UHF RFID and PLC integration. Video.
- **Bathymetric survey boat:** electronics and remote control for the Algerian National Hydrological Agency. Publication.

Additional platform experience: UR5, UFactory xArm6 and Lite6, SO-ARM100, MiR mobile robots and Enabled Robotics mobile manipulators.

Technical Expertise

- **Robot learning:** PyTorch; reinforcement learning (PPO, SAC, TD3, DQN, offline RL); multi-agent learning; imitation learning; world models; transformers and vision–language–action models; sim-to-real transfer.
- **Planning and coordination:** multi-agent path finding; multi-arm motion planning; task allocation and scheduling; collision avoidance; mixed-integer optimization (CPLEX), genetic algorithms and vehicle routing.
- **Control and mechatronics:** kinematics and dynamics; nonlinear, fuzzy, sliding-mode and CPG control; control barrier functions; embedded motor control; STM32, Arduino and PIC; robot and sensor integration.
- **Software and automation:** Python, C/C++, MATLAB/Simulink, Java, C# and R; ROS 1/2, MoveIt; MuJoCo, Isaac Lab, PyBullet and CoppeliaSim; Siemens TIA Portal, Ladder and Grafcet; Git, Docker and Linux.

Invited Talks and Research Presentations

- **Reinforcement Learning in Robotics: From Mathematical Foundations to Real-World Applications.** Invited seminar, Electrical Engineering Department, KFUPM, November 2025.
- **Decentralized Multi-Robot Motion Planning and Coordination for Industrial Applications.** Invited seminar, University of Tartu, Estonia, September 2024.
- **Multi-Robot Motion Planning and Coordination in Industrial Environments.** Speaker and co-organizer, Fourth Annual Research Workshop, KFUPM, August 2024.
- Guest lectures at SDU Biorobotics (2021–2023); research demonstrations at the KFUPM Research Forum (October 2024); presented research at several international conferences in robotics, automation and industrial informatics.

Professional Service and Industry Engagement

- **Editorial and reviewing:** Topic Editor, *Frontiers in Robotics and AI*, “Human-Robot Interaction in Industrial Settings: New Challenges and Opportunities”; reviewer for robotics and AI journals and conferences.
- **Research infrastructure and open science:** Robotics Laboratory coordination; shared manipulation testbed; open SPINDER hardware and control code, ANTS planner and lifelong multi-agent benchmark datasets.
- **Industrial collaboration:** robotics and automation projects with Siemens Algeria, Enabled Robotics, Mobile Industrial Robots, Sonacome and the Algerian National Hydrological Agency.

Professional References

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Publications

Author name shown in bold. Titles link to publisher records where available. Google Scholar.

Journal Articles — Published and Accepted

[J1] A. Aziz Khater, Y. Chen, **A. Maoudj**, A. Aouaichia, "Lyapunov-based adaptive path tracking control of wheeled mobile robots using a self-organizing recurrent Gaussian-gating functional-link fuzzy neural network," *Scientific Reports, published online 1 November 2026*.

[J2] B. Brahmi, **A. Maoudj**, J. Ghommam, "Nonlinear Terminal Model-Based Second-Order Sliding Mode Control for Robust Tracking in Robotic Exoskeletons," *International Journal of Robust and Nonlinear Control, Early View, 30 October 2026*.

[J3] **A. Maoudj**, B. Brahmi, A. A. Saputra, A. Aouaichia, M. Nouioua, N. Kubota, "Decentralized multi-drone gas source localization via belief-coupled MAPPO and barrier-filtered safety," *Scientific Reports*, accepted, 2026.

[J4] **A. Maoudj**, A. A. Saputra, B. Brahmi, M. Nouioua, "SPINDER: An open-source 18-DoF hexapod robot with hierarchical central pattern generator control and analytic inverse kinematics," *Scientific Reports*, vol. 16, art. 26147, 2026.

[J5] **A. Maoudj**, A. A. Saputra, B. Brahmi, N. Kubota, "A hierarchical perception-action framework for adaptive legged locomotion in unstructured environments," *IEEE Access*, vol. 14, pp. 74923–74932, 2026.

[J6] N. Djouambi, M. A. Yallese, M. Kaddeche, **A. Maoudj**, A. Khellaf, M. Nouioua, "Machine-Learning Surrogates for the Multi-Objective Optimisation of Cutting Parameters in Dry Turning of POM C GF25%," *The International Journal of Advanced Manufacturing Technology*, vol. 145, no. 5–6, pp. 3651–3670, 2026.

[J7] M. Nouioua, A. Laouissi, A. A. D. Sarhan, V. Swaminathan, **A. Maoudj**, H. M. Al-Qahtani, "AI-Driven Decision Support for Multi-Objective Optimization of Turning Parameters in Grey Cast Iron Machining," *Results in Engineering*, vol. 29, article 109805, 2026.

[J8] A. Mechta, A. Laouissi, E. Raouache, **A. Maoudj**, M. Nouioua, "Hybrid Numerical-Experimental Optimization of Resistance Spot Welding Parameters for Dissimilar S235/AISI 1075 Steel Sheets via Box–Behnken Response Surface Methodology," *JOM*, vol. 78, no. 11, pp. 10785–10805, 2026.

[J9] Z. Khan, **A. Maoudj**, S. Mekid, "From collaborative robots to Internet of Cobots: Innovations, applications, and emerging trends," *Journal of Ambient Intelligence and Smart Environments*, vol. 18, no. 1, pp. 41–67, 2026.

[J10] M. Boumediene, **A. Maoudj**, A. L. Christensen, "HM-DRL: Enhancing multi-agent pathfinding with a heatmap-based heuristic for distributed deep reinforcement learning," *Applied Intelligence*, vol. 55, no. 12, art. 873, 2025.

[J11] U. Farhadi, H. Hess, **A. Maoudj**, A. L. Christensen, "Evolution of path costs for efficient decentralized multi-agent pathfinding," *Swarm and Evolutionary Computation*, vol. 93, art. 101833, 2025.

[J12] **A. Maoudj**, A. L. Christensen, "Improved decentralized cooperative multi-agent path finding for robots with limited communication," *Swarm Intelligence*, vol. 18, no. 2–3, pp. 167–185, 2024.

[J13] A. Hentout, **A. Maoudj**, A. Kouider, "Shortest path planning and efficient fuzzy logic control of mobile robots in indoor static and dynamic environments," *Romanian Journal of Information Science and Technology*, vol. 27, no. 1, pp. 21–36, 2024.

[J14] **A. Maoudj**, A. Kouider, A. L. Christensen, "The capacitated multi-AGV scheduling problem with conflicting products: Model and a decentralized multi-agent approach," *Robotics and Computer-Integrated Manufacturing*, vol. 81, art. 102514, 2023.

[J15] A. Hentout, **A. Maoudj**, M. Aouache, "A review of the literature on fuzzy-logic approaches for collision-free path planning of manipulator robots," *Artificial Intelligence Review*, vol. 56, no. 4, pp. 3369–3444, 2023.

[J16] A. Hentout, **A. Maoudj**, A. Kouider, M. Aouache, "Effective GA approach for municipal solid waste collection: Dynamic capacitated vehicle routing in smart cities," *International Journal of Artificial Intelligence and Soft Computing*, vol. 7, no. 4, pp. 329–352, 2022.

[J17] **A. Maoudj**, A. Hentout, "Optimal path planning approach based on Q-learning algorithm for mobile robots," *Applied Soft Computing*, vol. 97, art. 106796, 2020.

[J18] A. Hentout, **A. Maoudj**, N. Kaid-Youcef, D. Hebib, B. Bouzouia, "Distributed multi-agent bidding-based approach for the collaborative mapping of unknown indoor environments by a homogeneous mobile robot team," *Journal of Intelligent Systems*, vol. 29, no. 1, pp. 84–99, 2020 (online 2017).

[J19] A. Hentout, M. Aouache, **A. Maoudj**, I. Akli, "Human-robot interaction in industrial collaborative robotics: A literature review of the decade 2008–2017," *Advanced Robotics*, vol. 33, no. 15–16, pp. 764–799, 2019.

[J20] **A. Maoudj**, B. Bouzouia, A. Hentout, A. Kouider, R. Toumi, "Distributed multi-agent scheduling and control system for robotic flexible assembly cells," *Journal of Intelligent Manufacturing*, vol. 30, no. 4, pp. 1629–1644, 2019.

[J21] A. Hentout, **A. Maoudj**, M. Ouahal, A. Gasmi, "Fault-tolerant multi-agent scheme for leader/follower formation control of homogeneous mobile robot team," *International Journal of Systems, Control and Communications*, vol. 10, no. 2, pp. 95–125, 2019.

[J22] A. Hentout, **A. Maoudj**, D. Guir, S. Saighi, M. A. Harkat, M. Z. Hammouche, A. Bakdi, "Collision-free path planning for indoor mobile robots based on rapidly-exploring random trees and piecewise cubic Hermite interpolating polynomial," *International Journal of Imaging and Robotics*, vol. 19, no. 3, 2019.

[J23] A. Hentout, **A. Maoudj**, D. Yahiaoui, M. Aouache, "RRT-A*-BT approach for optimal collision-free path planning for mobile robots," *Algerian Journal of Signals and Systems*, vol. 4, no. 2, pp. 39–50, 2019.

[J24] **A. Maoudj**, A. Hentout, B. Bouzouia, R. Toumi, "On-line fault-tolerant fuzzy-based path planning and obstacles avoidance approach for manipulator robots," *International Journal of Uncertainty, Fuzziness and Knowledge-Based Systems*, vol. 26, no. 5, pp. 809–838, 2018.

[J25] A. Bakdi, A. Hentout, H. Boutami, **A. Maoudj**, O. Hachour, B. Bouzouia, "Optimal path planning and execution for mobile robots using genetic algorithm and adaptive fuzzy-logic control," *Robotics and Autonomous Systems*, vol. 89, pp. 95–109, 2017.

Refereed Conference Papers

[C1] **A. Maoudj**, A. Kharidege, B. Brahmi, A. A. Saputra, M. Nouioua, "An Adaptive Vision-Based Robotic System for Flexible Surface Polishing," *23rd International Conference on Informatics in Control, Automation and Robotics (ICINCO)*, vol. 2, pp. 1091–1098, 2026.

[C2] B. Brahmi, **A. Maoudj**, J. Ghommam, M. H. Rahman, "Hamiltonian-Based Human Intention Estimation for Compliant Exoskeleton Robot," *23rd International Conference on Informatics in Control, Automation and Robotics (ICINCO)*, vol. 1, pp. 495–502, 2026.

[C3] K. Amara, W. Dib, A. O. Djekoune, **A. Maoudj**, N. Zenati, "AquaRob: An unmanned surface drone for the coastal surveillance and environmental monitoring," *Proc. 4th International Conference on Embedded & Distributed Systems (EDiS), IEEE*, pp. 193–198, 2024.

[C4] **A. Maoudj**, S. Mekid, A. L. Christensen, "Lifelong MAPF for shipment preparation in stochastic warehouses," *Proc. IEEE 20th International Conference on Automation Science and Engineering (CASE)*, pp. 1006–1011, 2024.

[C5] **A. Maoudj**, A. L. Christensen, "Decentralized multi-agent path finding in dynamic warehouse environments," *Proc. 21st International Conference on Advanced Robotics (ICAR), IEEE*, pp. 28–34, 2023.

[C6] **A. Maoudj**, A. L. Christensen, "Decentralized multi-agent path finding in warehouse environments for fleets of mobile robots with limited communication range," *Swarm Intelligence: 13th International Conference (ANTS 2022), Springer LNCS vol. 13491*, pp. 104–116, 2022.

[C7] M. Gaham, M. Kadri, **A. Maoudj**, A. Bendjelloul, A. Kouider, B. Mihoubi, K. Tebani, B. Bouzouia, "Multi-agent control of an Industry 4.0 cyber-physical assembly cell," *1st National Conference on Applied Computing and Smart Technologies, Sidi Bel Abbès, Algeria*, 2021.

[C8] **A. Maoudj**, A. Hentout, A. L. Christensen, A. Kouider, "A fast constructive path planning algorithm for mobile robot navigation," *Proc. 26th IEEE International Conference on Emerging Technologies and Factory Automation (ETFA)*, pp. 1–8, 2021.

[C9] **A. Maoudj**, A. L. Christensen, "Q-learning-based navigation for mobile robots in continuous and dynamic environments," *Proc. IEEE 17th International Conference on Automation Science and Engineering (CASE)*, pp. 1338–1345, 2021.

[C10] **A. Maoudj**, A. Hentout, M. Bouhebila, C. Boubakir, H. Maoudj, "Design and control of an educational and research-oriented mobile robot," *International Conference on Networking, Telecommunications, Biomedical Engineering and Applications (ICNTBA)*, 2019.

[C11] H. Maoudj, **A. Maoudj**, A. Hentout, M. Bouhebila, C. Boubakir, "Autonomous fuzzy-based navigation of mobile robots: Design, realization and control of a prototype," *International Conference on Networking, Telecommunications, Biomedical Engineering and Applications (ICNTBA)*, 2019.

[C12] A. Hentout, A. Tiberkak, **A. Maoudj**, L. Berkat, A. Abeb, "Virtual pheromone-based approach for objects searching in RFID-based cyber-physical robotic systems," *Proc. International Conference on Applied Smart Systems (ICASS)*, pp. 1–7, 2018.

[C13] A. Hentout, A. Beghni, A. B. Nourine, **A. Maoudj**, B. Bouzouia, "Mobile robot RBPF-SLAM with LMS sensor in indoor environments: Application on RobuTER robot," *Proc. 5th International Conference on Electrical Engineering, Boumerdes (ICEE-B), IEEE*, 2017.

[C14] A. Hentout, **A. Maoudj**, B. Bouzouia, "A survey of development frameworks for robotics," *Proc. 8th International Conference on Modelling, Identification and Control (ICMIC)*, pp. 67–72, 2016.

[C15] **A. Maoudj**, B. Bouzouia, A. Hentout, A. Kouider, R. Toumi, "Distributed multi-agent approach based on priority rules and genetic algorithm for tasks scheduling in multi-robot cells," *Proc. IECON 2016, 42nd Annual Conference of the IEEE Industrial Electronics Society*, pp. 692–697, 2016.

[C16] **A. Maoudj**, B. Bouzouia, A. Hentout, R. Toumi, "Multi-agent approach for task allocation and scheduling in cooperative heterogeneous multi-robot team: Simulation results," *Proc. IEEE 13th International Conference on Industrial Informatics (INDIN)*, pp. 179–184, 2015.

[C17] **A. Maoudj**, B. Bouzouia, A. Hentout, R. Toumi, "Multi-agent architecture for telerobotics of heterogeneous multi-robot cooperation through negotiated task allocation with constraints: Initial implementation," *1re Journée d'Étude en Automatique et ses Applications (1JEAA14)*, pp. 38–43, 2014.

Editorial, Workshop Paper, Preprints and Theses

[O1] S. Bouraine, M. Ammi, K. Geihs, A. Hentout, **A. Maoudj**, F. Yacef, "Editorial: Human-robot interaction in industrial settings: new challenges and opportunities," *Frontiers in Robotics and AI*, vol. 12, art. 1652426, 2025.

[O2] A. Hentout, M. Aouache, **A. Maoudj**, I. Akli, "Key challenges and open issues of industrial collaborative robotics," *Workshop on Human-Robot Interaction: From Service to Industry (HRI-SI), held at IEEE RO-MAN*, 2018.

[O3] A. Kharidege, **A. Maoudj**, "Integration of a vision system with a robotic polishing system for machining variety of workpieces," *Research Square preprint*, 2024.

[O4] A. Hentout, **A. Maoudj**, A. Kouider, "Efficient fuzzy-logic control of indoor mobile robots evolving in static and dynamic workspaces," *Research Square preprint*, 2023.

[O5] A. Hentout, **A. Maoudj**, A. Kouider, "A literature review of service robotics: Challenges and open research issues," *Workshop on Human-Robot Interaction: From Service to Industry (HRI-SI)*, 2022.

[O6] A. Hentout, I. Akli, **A. Maoudj**, A. Tiberkak, "Developing a distributed architecture for robots control in industrial RFID-based cyber-physical systems," *technical report/preprint*, 2018.

[O7] **A. Maoudj**, "Téléopération multi-robots hétérogènes : Application au cas des robots RobuTER/ULM et AGV-M6," *Doctoral thesis, USTHB*, 2018.

[O8] **A. Maoudj**, C. E. Boubakir, "Conception et commande d'un prototype mobile à base de microcontrôleur," *Master's thesis, Mohammed-Seddik Benyahia University of Jijel*, 2012.

Journal Manuscripts — Under Review and Submitted

[U1] A. A. Saputra, **A. Maoudj**, M. L. Afakh, A. R. Ahmad, S. Mekid, N. Kubota, "Decentralized RL-CPG control for hexapod locomotion under simultaneous multi-leg failures," *IEEE Transactions on Robotics*, under review, 2026.

[U2] **A. Maoudj**, S. Mekid, M. Nouioua, "ML-BiRRT: Machine-learning-enhanced bi-directional RRT for multi-arm robot motion planning and coordination," *Robotics and Autonomous Systems*, under review, 2026.

[U3] **A. Maoudj**, A. A. Khater, B. Brahmi, A. Aouaichia, A. M. Sallam, A. A. Saputra, "Online fuzzy-neural actor-critic control for hexapod trajectory tracking," *Applied Soft Computing*, under review, 2026.

[U4] M. Boumediene, A. L. Christensen, **A. Maoudj**, "MAPF-Relocate: Mitigating congestion through strategic agent relocation in multi-agent pathfinding," *Journal of Intelligent & Robotic Systems*, under review, 2026.

[U5] **A. Maoudj**, A. A. Khater, B. Brahmi, A. Aouaichia, A. M. Sallam, A. A. Saputra, P. K. M. Parameswarappa, "Lyapunov-stable path tracking control of wheeled mobile robots using self-organizing recurrent probabilistic functional-link fuzzy neural network," *Biomimetic Intelligence and Robotics*, under review, 2026.

[U6] B. Brahmi, **A. Maoudj**, N. Iqbal, M. H. Rahman, "Model-based terminal second-order sliding mode control with exact finite-time convergence for fault-tolerant exoskeleton robots," *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, under review, 2026.

[U7] A. A. Khater, A. Aouaichia, **A. Maoudj**, Y. Chen, "An advanced recurrent probabilistic TSK neuro-fuzzy identifier for mobile robot system," *Engineering Applications of Artificial Intelligence*, under review, 2026.

[U8] S. Alqahtani, **A. Maoudj**, S. Bashmal, "Learning-based kinematic modeling and disturbance compensation for a 3UPU parallel kinematic manipulator under base vibrations," *submitted*, 2026.